VICTOR IBHAFIDON

Software Engineer 3

London, United Kingdom | victoribhafidon@outlook.com | 07425 871413 | linkedin.com/in/victoribhafidon | github.com/Web8080

PROFESSIONAL SUMMARY

My approach to software engineering is grounded in the belief that good software is built through thoughtful design, sound engineering, and continuous iteration. I've worked on AI/ML solutions across software engineering, data, security, and cloud, gaining experience through building, iterating, and learning from challenges along the way, while balancing technical requirements with broader business needs. My philosophy is simple: I don't just want something that works in a demo. I want to understand what can go wrong, prove it works, and build it properly. I work across the full stack, including React and TypeScript interfaces backed by the Python services I build.

TECHNICAL SKILLS

Programming: Python, TypeScript, SQL, JavaScript, Bash

Backend & Data: Redis, FastAPI, Django, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Neo4j, pgvector, Vector Search, Data Pipelines, ETL

Cloud & DevOps: CI/CD, AWS (EC2, S3, Lambda, IAM, EKS), Microsoft Azure, Azure OpenAI, Docker, Kubernetes, GitHub Actions, Linux, Nginx, OpenTelemetry, Prometheus, Grafana

AI & Machine Learning: LLMs, Generative AI, AI Agents, Multi-Agent Systems, LangGraph, MCP, JSON-RPC, Tool Calling, RAG, GraphRAG, Prompt Engineering, AI Evaluation, PyTorch, Transformers, Scikit-learn, Computer Vision, NLP

Security & AI Governance: AI Security, AI Governance, Authentication, Authorization, RBAC, API Security, OWASP Top 10, Prompt Injection Mitigation, AI Safety, Responsible AI, GDPR

MLOps & Model Operations: MLflow, Model Training, Model Evaluation, Model Monitoring, Drift Detection, Experiment Tracking, Model Deployment

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML), Bastion Shield Technologies

London, United Kingdom | April 2026 – Present

- Built Bulwark, a multi-tenant AI agent governance platform that gives organisations control and visibility over AI agents. Designed and implemented agent identity and lifecycle management, policy enforcement, human oversight workflows, and detection of unregistered agents, orphaned ownership, and stale credentials.
- Engineered the backend in Python, TypeScript, and PostgreSQL, implementing RBAC, server-side tenant isolation, secure authentication, API-key management, and an SDK for external teams to register agents and stream policy decisions. Designed tamper-evident audit logging with offline verification, strengthened data privacy controls, and remediated four production vulnerabilities, including a session-token forgery path.
- Owned the cloud infrastructure using Terraform-managed AWS environments across development, staging, and production. Containerised the platform with Docker and delivered it through GitHub Actions CI/CD, while working on distributed system patterns including queues, event streaming, caching, and horizontal scaling to improve performance and latency.
- Developed behavioural anomaly detection combining statistical methods with usage forecasting to identify unusual agent activity. The system achieved a 0.981 ROC-AUC against an Isolation Forest baseline and was integrated into the TypeScript runtime for real-time scoring.

Full Stack Software Engineer, Basketball Nxtion

London, United Kingdom | February 2026 – April 2026

- Developed the MVP of Plxyground, a mobile application connecting basketball players, creators, and sports communities, contributing across product design, application architecture, backend development, APIs, authentication, and user workflows.
- Designed the PostgreSQL data architecture and analytics pipelines for the MVP, structuring user, activity, engagement, and platform data to support data ingestion, transformation, analysis, and business intelligence. Built data workflows enabling insights into platform performance, user behaviour, engagement, and product adoption.
- Designed and integrated an AI architecture connecting LLM/ML services with PostgreSQL data, backend APIs, and application business logic. Built workflows for personalised recommendations, intelligent user matching, and content discovery, using user and platform data to generate relevant results, improve engagement, and automate discovery workflows.
- Deployed and maintained the MVP infrastructure on AWS, implementing Docker, CI/CD, testing, rollbacks, monitoring, and third-party integrations while collaborating with frontend engineers and stakeholders throughout the software development lifecycle from requirements and development through testing, deployment, and iteration.

Full Stack Python Engineer Intern, CyberAI Technologies

London, United Kingdom | September 2025 – November 2025

- Collaborated with senior AI engineers to develop Surfyn AI, a conversational AI customer support platform enabling businesses to deploy knowledge-based AI assistants grounded in websites and documents to automate customer interactions and provide relevant, context-aware responses.
- Contributed to the development of RAG and conversational AI workflows using Python and Django, working on document ingestion, chunking, embeddings, vector-based retrieval, prompt construction, conversation context, REST APIs, and LLM integrations while learning how knowledge-based AI systems are designed and evaluated.
- Developed backend functionality and AI workflow improvements, working on API development, debugging, testing, performance optimisation, knowledge retrieval, and integration between AI services and the Django application.
- Worked with the AI Security Engineering team on AI evaluation, testing, safety, and deployment workflows, helping investigate retrieval quality, response relevance, hallucination risks, prompt injection, sensitive data handling, logging, and application reliability across production AI systems.

SELECTED PROJECTS

Portivo Estate Suite: Took ownership of the system design and development of an AI powered property operations platform for UK letting agents and BTR operators, covering lettings, sales, maintenance and compliance in one system. Built Portivo end to end with a React, TypeScript and Tailwind frontend coupled with a Python and FastAPI backend using SQLAlchemy and PostgreSQL for the multi-tenant data model. Implemented Alembic schema migrations against a live customer database, Redis and Celery for caching and scheduled background jobs, tenant isolation at the query layer, and role-based access control at the endpoint level. The AI layer uses a LangGraph multi-agent workflow on Claude models, with an orchestrator delegating typed tasks to specialist agents for document retrieval, lease abstraction, arrears detection, compliance tracking, maintenance triage and outbound sales prospecting. Built an OCR, chunking and embedding pipeline using pgvector with per-tenant vector scoping and source-referenced answers, while agents access CRM, email and document capabilities through MCP servers with scoped credentials rather than bespoke integrations. Evaluated against labelled datasets for groundedness and extraction accuracy, traced with Langfuse, and guarded using schema-validated outputs and confidence thresholds. Deployed to the cloud using Docker and GitHub Actions CI/CD, with an audit trail covering every agent action. The interface was built in React, TypeScript and Tailwind directly against the FastAPI backend, covering the full path from data model through API to the screens users work in. Live at portivoestatesuite.com

Automated Drone Security & Inspection System: Designed an intelligent drone security system that protects high-value infrastructure across the UK — railway lines, power stations, solar and wind farms, ports, reservoirs, and industrial sites. The system uses autonomous drones that follow pre-set patrol routes or respond to security alerts. Smart cameras automatically detect threats: trespassers, unauthorized access, overgrown vegetation near equipment, and mechanical faults. Security operators monitor everything through a live dashboard showing drone locations, real-time video feeds, and prioritized alerts. Every drone deployment requires human approval before launch — ensuring safety and accountability at all times. Technical Implementation: Autonomous drone patrol and anomaly detection for critical infrastructure: rail corridors, electrical substations, solar and wind farms, ports, reservoirs and industrial perimeters. Drones fly scheduled or operator-triggered missions. Onboard computer vision detects intrusions, trespass, vegetation encroachment and equipment faults. Operators get a live map, live video, an alert queue and a full audit trail. Every dispatch is approved by a human before the aircraft arms. Built real-time telemetry and event-driven processing with Kafka, FastAPI, PostgreSQL and Redis, integrating YOLO and PyTorch for computer vision inference, training and fine-tuning, with MLOps for model evaluation, drift detection and retraining. Designed bounded autonomous decision-making and control workflows with operator override, emergency-stop and failure-mode handling, alongside robotics interfaces for drones and ground vehicles using MQTT and ROS. Implemented AI security controls including OWASP-aligned protections, RBAC, JWT authentication, audit logging, SSRF and injection protections. Containerised services with Docker and designed Kubernetes, Terraform and AWS infrastructure for scalable deployment. Added real-time detection and alerting with streaming telemetry and WebSockets, with Prometheus and Grafana for observability and monitoring. GitHub: github.com/Web8080/Autonomous_AI_defense_system_design

Medical Imaging AI: Designed the data-preparation and augmentation pipeline and iteratively trained PyTorch CNN models across a 231,444-image dataset, using classification metrics and failure analysis to guide improvements up to 92% classification accuracy. Converted the trained model into a versioned, Docker-deployed inference service with health checks, audit trails and runtime monitoring.

Autonomous Energy Grid & Smart Battery Management: Built an end-to-end ML/MLOps system for battery-optimisation and energy-management strategy: turned raw energy and battery data into features for predictive modelling and anomaly detection, then evaluated candidate models against validation performance and operational requirements to balance accuracy, cost and reliability. Deployed versioned models via MLflow, Docker and Kubernetes with Prometheus and Grafana monitoring and drift detection, so model behaviour stays measurable after deployment.

EDUCATION

MSc Data Science, Nottingham Trent University, United Kingdom (2023 – 2024)

BSc Computer Science, Ambrose Alli University, Nigeria (2016 – 2021)